

Experiment : Depth First Search (DFS)

Aim

To write a Python program to implement Depth First Search (DFS) traversal on a graph.

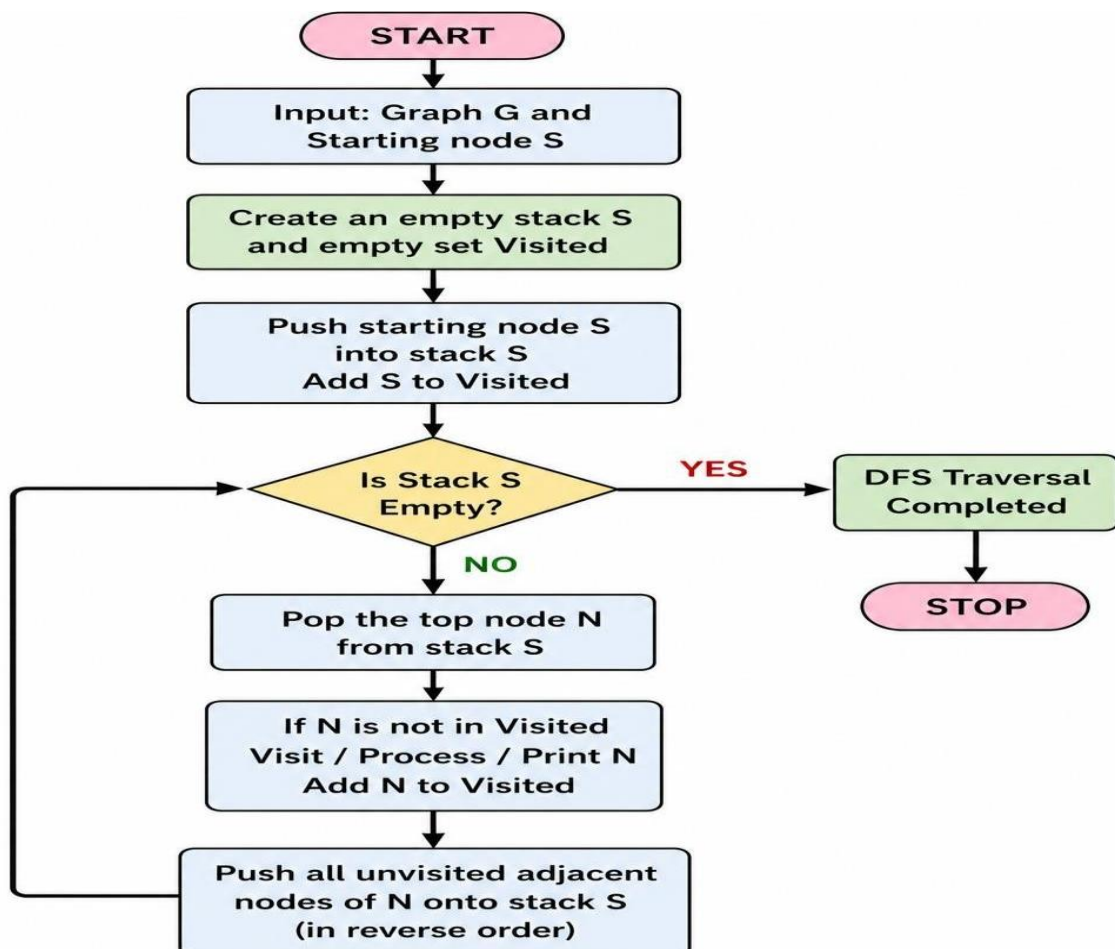
Objective

To traverse all the nodes of a graph using the DFS algorithm and understand the working of stack/recursion-based graph traversal.

Algorithm

- Create a graph using an adjacency list.
- Define a DFS function with the current node as input.
- Mark the current node as visited.
- Print the current node.
- Recursively visit all unvisited adjacent nodes.
- Repeat until all reachable nodes are visited.

Flowchart:



Code

Python Run

```
g = {'A':['B','C'], 'B':['D'], 'C':['E'], 'D':[], 'E':[]}

v = set()

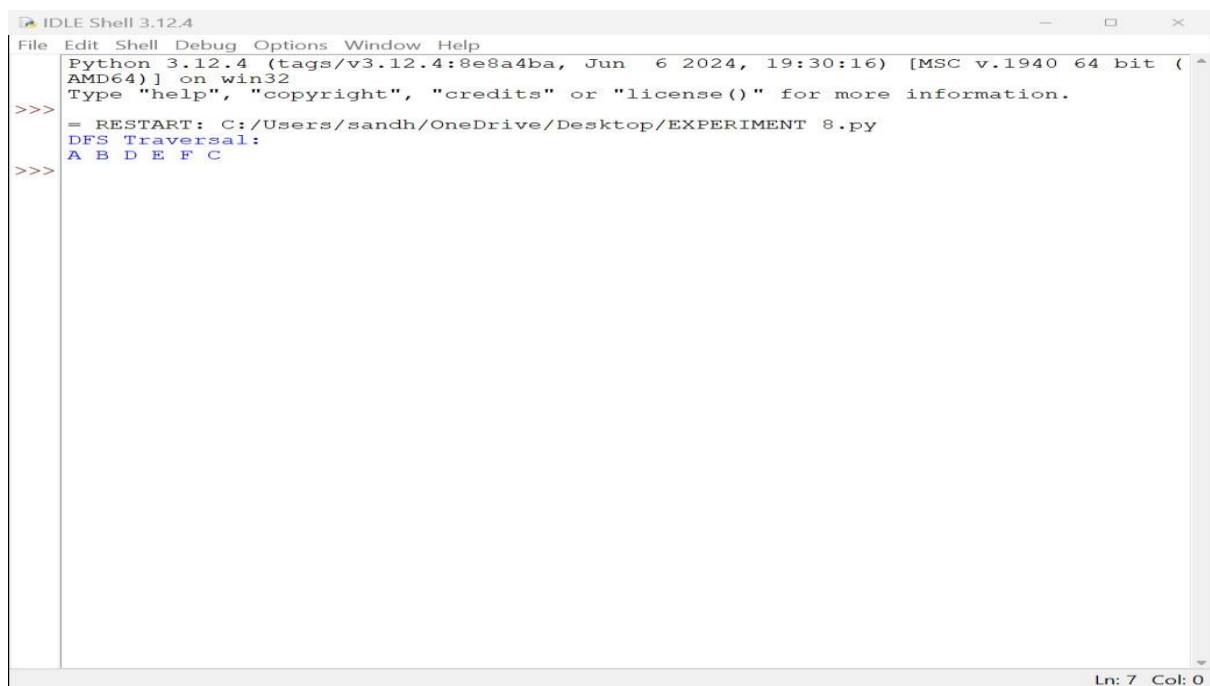
def dfs(n):
    if n not
in v:
    print(n,
end=' ')

    v.add(n)

    for i in g[n]:
        dfs(i)

print("DFS Traversal:") dfs('A')
```

Output



```
IDLE Shell 3.12.4
File Edit Shell Debug Options Window Help
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> = RESTART: C:/Users/sandh/OneDrive/Desktop/EXPERIMENT 8.py
DFS Traversal:
A B D E F C
>>>
```

Result

The DFS traversal of the given graph was successfully performed using Python, and the nodes were visited in the order: **A → B → D → C → E**.

Conclusion

The Depth First Search algorithm was implemented successfully in Python. DFS explores nodes deeply before backtracking and is useful for path finding, cycle detection, and graph-based applications.